S3.1 The periodic table: Classification of elements

The big picture

Guiding question(s)

How does the periodic table help us to predict patterns and trends in the properties of elements?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Development of the periodic table

Russian chemist Dmitri Mendeleev is credited with developing the model of the periodic table (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) that we are most familiar with today, the one used in this course and that you can find in your DP data booklet. While Mendeleev's periodic table looked a bit different than the one we are most familiar with, he developed the framework for our current model. This framework organised elements by increasing atomic mass and grouping elements with similar properties together. What made Mendeleev's organisation of elements unique was leaving blank spots in his model indicating an element with specific properties that had yet to be discovered. **Figure 1** shows a version of how Mendeleev organised elements.

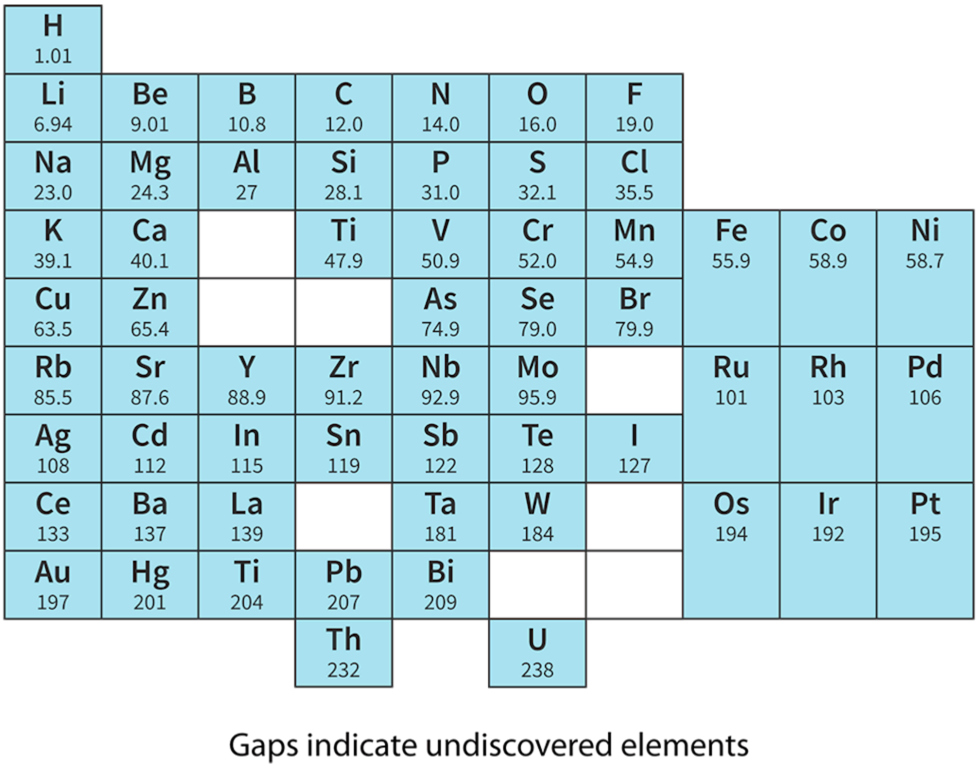


Figure 1. Mendeleev's organisation of the elements.

More information for figure 1

Each of the blank spots in **Figure 1** indicate an element that was not known at the time but has since been discovered. The first to be discovered was gallium. Based on the organisation of the elements, Mendeleev was able to accurately predict the mass, melting point and density of gallium before it was discovered. The ability to use the arrangement of the elements to make predictions is at the core of the subject of chemistry, even today.

Nature of Science

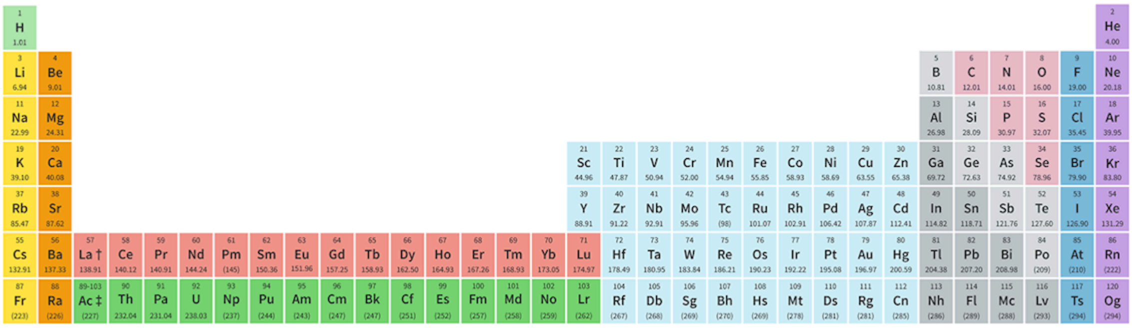
Aspect: Science as a shared endeavour

French chemist Paul-Émile Lecoq de Boisbaudran was the first to discover and isolate gallium, Ga. Mendeleev not only tried to claim credit for this discovery since he had predicted its existence, but he also refuted the quantitative properties Lecoq de Boisbaudran reported as they did not match his predictions. As Lecoq de Boisbaudran investigated further, it turned out his values for atomic mass and density were not correct and he had to publish a retraction.

While these two chemists each wanted sole credit for this discovery, pursuing accuracy, replicability and production of materials in science is a shared endeavour. Advancing our scientific understanding in chemistry is possible through research findings of both theorists and experimentalists and benefits from collaborative efforts.

Various models of the periodic table

While we are likely to be familiar with the periodic table used in this course, it is important to note that it is not the only organisation method to arrange the elements that exists. The model we use today is most commonly used because it is easy to interpret and can fit on a standard piece of paper. However, in order to make it fit easily on a standard piece of paper it does have two rows cut out of it and placed on the bottom, these two rows are called the lanthanide and actinide series. **Figure 2** shows a comparison of the periodic table with and without these series removed.



More information

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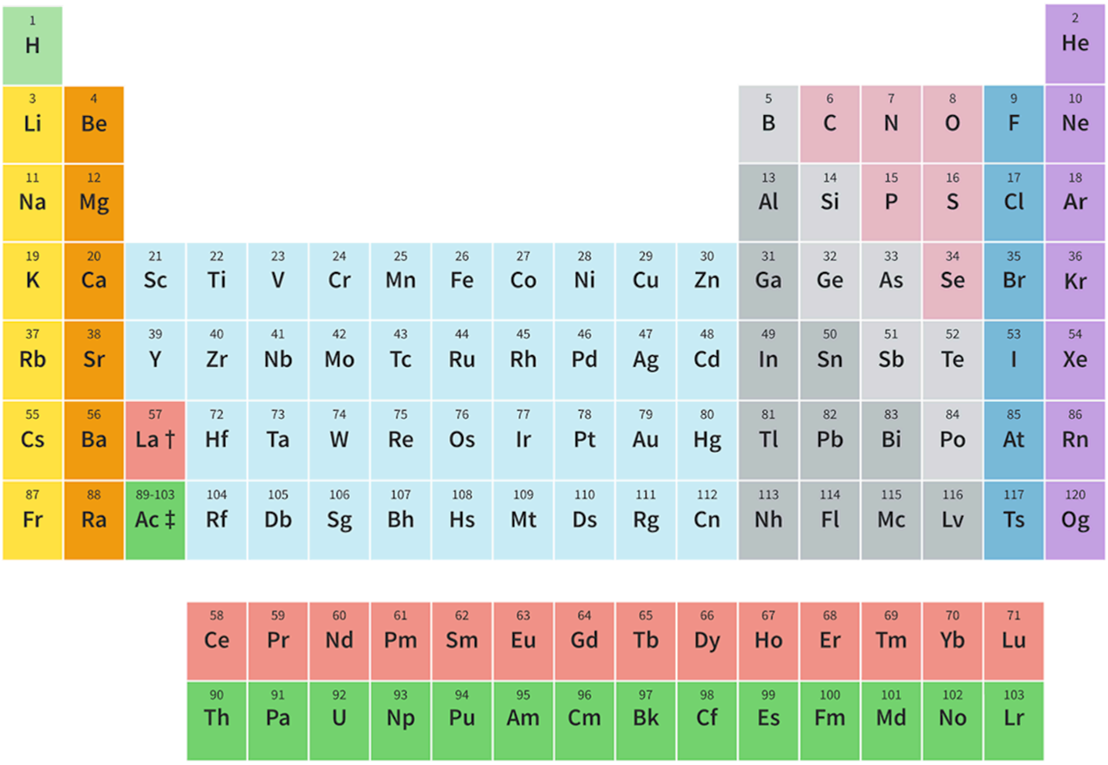

Figure 2. Two versions of the modern model of the periodic table.

More information for figure 2

There are many other models of the periodic table that each have their own strengths and weaknesses. Use **Interactive 1** to look at four of these examples and to examine some of the strengths and weaknesses of each. As you look through these, consider what situations you might find each of these models useful.

Interactive 1. Various Models of the Periodic Table.

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More information for interactive 1

Prior learning

Before you study this subtopic make sure that you understand the following:

- How to deduce the number of electrons in each energy level for an element (see section S1.3.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/)).
- How to deduce the electron configuration of an atom up to Z = 36 (see section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/)).
- Classifying elements as metals and non-metals (see section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/)).

Practical skills

Once you have completed this subtopic, apply your knowledge of the periodic table by going to Practical 3a: Exploring databases (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/exploring-databases-3a-id-46716/) and Practical 3b: Periodic trends database investigation (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodic-trends-database-investigation-3b-id-46717/).